

Power BI Basics Assignment – Detailed Solutions

Question 1 : What is Power BI and why is it used in businesses?

Power BI is a Business Intelligence (BI) and Data Visualization tool developed by Microsoft. It is used to collect, transform, analyze, and visualize data from different sources in the form of reports and dashboards.

Power BI helps businesses convert raw data into meaningful insights so that management can make informed decisions.

Why businesses use Power BI:

1. Data Visualization

- Power BI converts complex data into charts, graphs, maps, KPIs, and dashboards.
- This makes data easier to understand.

2. Better Decision Making

- Businesses can analyze trends, sales performance, customer behavior, and financial data.
- Managers can make faster and smarter decisions.

3. Real-Time Monitoring

- Power BI supports real-time dashboards.
- Businesses can track live performance metrics instantly.

4. Integration with Multiple Sources

- It can connect with Excel, SQL Server, Oracle, MySQL, Google Analytics, Azure, and many more.

5. Data Cleaning and Transformation

- Power Query allows cleaning and shaping data before analysis.

6. Interactive Reports

- Users can filter, drill down, and interact with reports dynamically.

7. Automation

- Scheduled refreshes automatically update reports with new data.

8. Cloud Accessibility

- Reports can be shared online using Power BI Service.

Example:

A retail company can use Power BI to analyze:

- Total sales
- Best-selling products
- Monthly revenue
- Regional performance
- Customer purchasing trends

Question 2 : Name and explain the three main components of Power BI.

The three main components of Power BI are:

1. Power BI Desktop

Power BI Desktop is a Windows application used for:

- Connecting to data sources
- Cleaning and transforming data
- Creating reports and dashboards
- Building data models

Features:

- Drag-and-drop report creation
- Power Query Editor
- DAX calculations
- Relationship management

2. Power BI Service

Power BI Service is a cloud-based online platform.

Functions:

- Publish reports online
- Share dashboards with users
- Schedule automatic data refresh
- Collaborate with teams

3. Power BI Mobile

Power BI Mobile is the mobile application available for Android and iOS.

Functions:

- View dashboards on mobile devices

- Access reports anywhere
- Receive notifications and alerts

Together, these three components create a complete Business Intelligence ecosystem.

Question 3 : Explain the Power BI workflow.

The Power BI workflow explains the complete process of converting raw data into meaningful business insights.

Step 1: Connect to Data Sources

Power BI connects with:

- Excel
- SQL databases
- CSV files
- Cloud services
- APIs

Step 2: Data Cleaning and Transformation

Performed using Power Query.

Tasks include:

- Removing duplicates
- Handling missing values
- Splitting columns
- Changing data types
- Filtering unnecessary data

Step 3: Data Modeling

Relationships are created between tables.

Activities:

- Creating relationships
- Defining primary and foreign keys
- Creating calculated columns
- Writing DAX formulas

Step 4: Report Creation

Visualizations are created such as:

- Bar charts
- Pie charts

- Tables

- KPIs

- Maps

Step 5: Dashboard Creation

Multiple report visuals are combined into dashboards.

Purpose:

To provide a summarized business overview.

Step 6: Publish Reports

Reports are published to Power BI Service.

Step 7: Sharing and Collaboration

Reports are shared with:

- Managers

- Teams

- Stakeholders

Step 8: Scheduled Refresh

Power BI automatically refreshes data periodically.

Question 4 : List any four data cleaning tasks that can be performed in Power Query.

Power Query is used to clean and transform raw data before analysis.

Four important data cleaning tasks are:

1. Remove Duplicate Records

Duplicate rows can be removed to avoid repeated values.

2. Handle Missing Values

Null or blank values can be:

- Replaced

- Removed

- Filled with default values

3. Change Data Types

Columns can be converted into:

- Text
- Number
- Date
- Currency

4. Split or Merge Columns

A column can be divided into multiple parts.

Additional tasks:

- Rename columns
- Filter rows
- Remove errors
- Format text
- Trim spaces
- Replace values

Question 5 : Write step by step instructions to load the above dataset.

Steps to load a dataset into Power BI:

Step 1:

Open Power BI Desktop.

Step 2:

Click on Home → Get Data.

Step 3:

Select the data source type:

- Excel
 - CSV
 - SQL Server
 - Text file
- etc.

Step 4:

Browse and select the dataset file.

Step 5:

Click Open.

Step 6:

A Navigator window will appear showing available tables/sheets.

Step 7:

Select the required table or sheet.

Step 8:

Click:

- Load → directly load data

OR

- Transform Data → clean data before loading

Step 9:

If Transform Data is selected:

- Open Power Query Editor

- Perform cleaning tasks

- Click Close & Apply

Step 10:

The dataset will load into Power BI.

Step 11:

Open Report View to create visualizations.

Question 6 : Define Data View, Report View, and Model View. Explain the purpose of each view.

1. Data View

Data View displays data in tabular form.

Purpose:

- View imported data

- Check records and columns

- Create calculated columns

- Verify cleaned data

Example:

Viewing customer or sales tables directly.

2. Report View

Report View is used to create visual reports and dashboards.

Purpose:

- Create charts and graphs
- Build KPIs
- Design dashboards
- Add filters and slicers

Example:

Creating a sales performance dashboard.

3. Model View

Model View shows relationships between tables.

Purpose:

- Create table relationships
- Manage primary and foreign keys
- Design data models
- Optimize data structure

Importance:

All three views work together for complete data analysis and reporting.

Question 7 : Discuss the different data sources that Power BI supports.

Power BI supports a large variety of data sources.

1. File-Based Sources

Examples:

- Excel
- CSV
- XML
- JSON
- PDF

2. Database Sources

Examples:

- SQL Server
- MySQL
- Oracle
- PostgreSQL

- MongoDB

3. Cloud-Based Sources

Examples:

- Azure
- Google Analytics
- Salesforce
- SharePoint
- OneDrive

4. Web and API Sources

Examples:

- Web pages
- REST APIs

5. Big Data Sources

Examples:

- Hadoop
- Spark

6. Power Platform Sources

Examples:

- Power Apps
- Power Automate.

Benefits of multiple data source support:

- Centralized reporting
- Unified analytics
- Real-time insights
- Better business intelligence

Question 8 : Split Owner Name to create two new columns as First Name and Last Name

Steps to split Owner Name column in Power Query:

Step 1:

Load the Real_Estate dataset into Power BI.

Step 2:

Click Transform Data to open Power Query Editor.

Step 3:

Select the Owner Name column.

Step 4:

Go to Transform tab.

Step 5:

Click Split Column.

Step 6:

Choose:

Split by Delimiter.

Step 7:

Select Space as the delimiter.

Step 8:

Choose:

Split at Left-most delimiter.

Step 9:

Click OK.

Step 10:

Two new columns will be created:

- First Name

- Last Name

Example:

Original Value:

Rahul Sharma

After Splitting:

First Name = Rahul

Last Name = Sharma